

Classical Conditioning Pt 2

Learning and Behavior (PSY/NSC 3320)

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Baylor University

Theoretical Explanations of Classical Conditioning

Theoretical Explanations of Classical Conditioning

Stimulus-Substitution Theory

Preparatory-Response Model

Rescorla-Wagner Theory

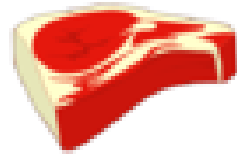
+ Opponent Process Theory

Theoretical Explanations of **Classical Conditioning**

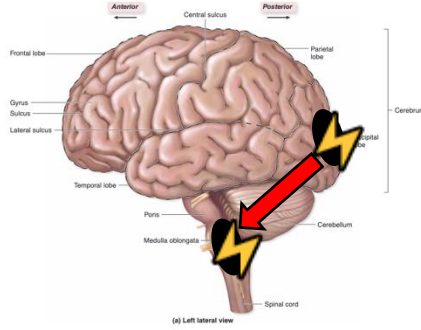
Stimulus-Substitution Theory – CS acts as substitute for US in triggering neural activity



Unconditioned Response
(Salivation)



Unconditioned Stimulus
(Food)



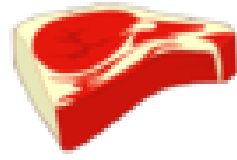
Salivation – Medulla

Theoretical Explanations of Classical Conditioning

Stimulus-Substitution Theory – CS acts as substitute for US in triggering neural activity



Unconditioned Response
(Salivation)



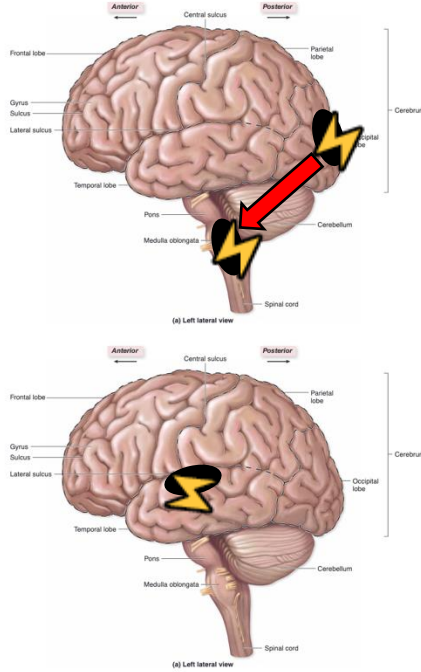
Unconditioned Stimulus
(Food)



No Response



Neutral Stimulus
(Bell Ringing)



Theoretical Explanations of Classical Conditioning

Stimulus-Substitution Theory – CS acts as substitute for US in triggering neural activity

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Stimulus-Substitution Theory – CS acts as substitute for US in triggering neural activity



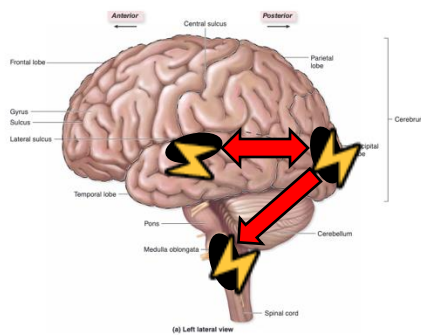
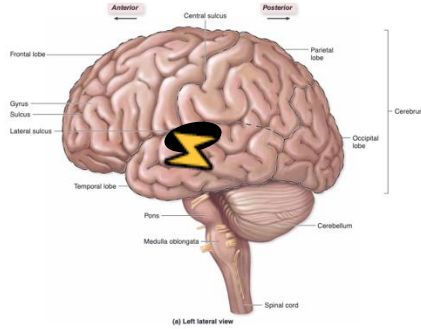
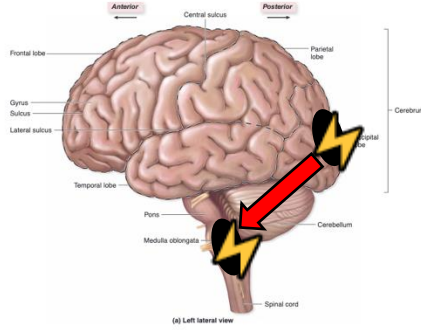
Neutral Stimulus (Bell Ringing)



Neutral Stimulus (Bell Ringing)



Unconditioned Stimulus (Food)



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Neutral Stimulus (Bell Ringing)



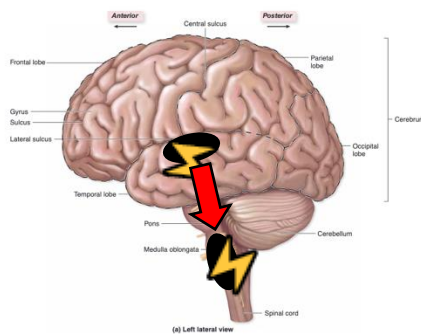
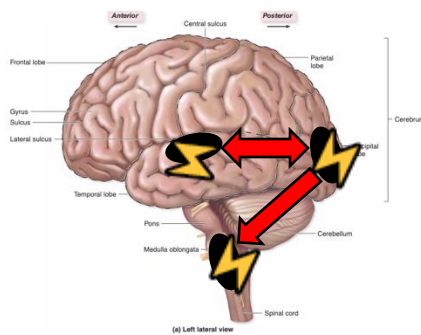
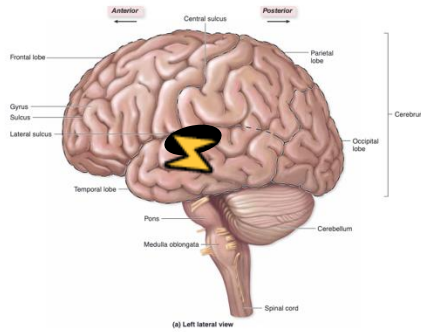
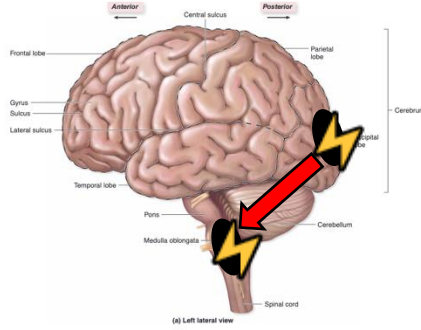
Neutral Stimulus (Bell Ringing)



Unconditioned Stimulus (Food)

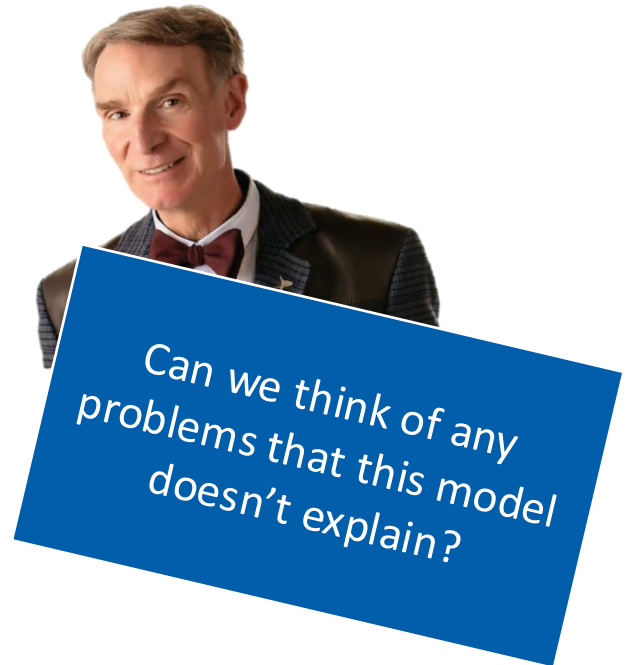
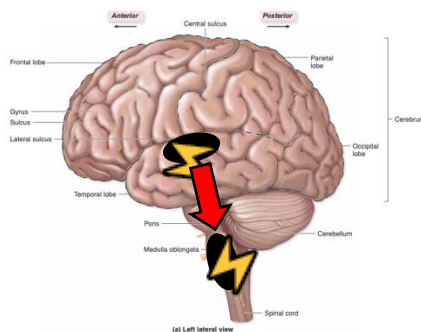
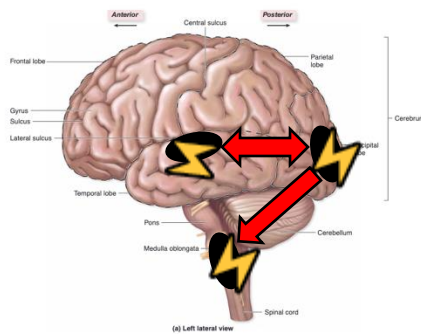
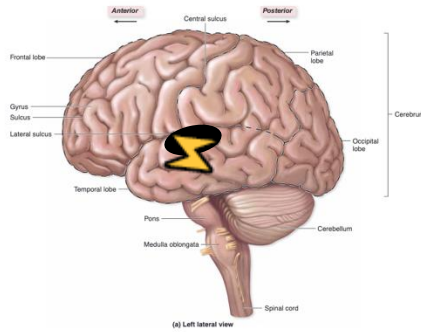
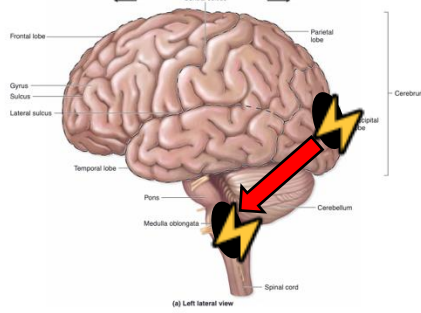
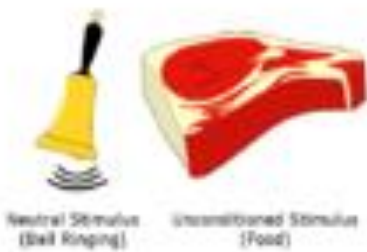


Conditioned Stimulus (Bell Ringing)



Theoretical Explanations of Classical Conditioning

Stimulus-Substitution Theory – CS acts as substitute for US in triggering neural activity



Theoretical Explanations of Classical Conditioning

Stimulus-Substitution Theory – CS acts as substitute for US in triggering neural activity

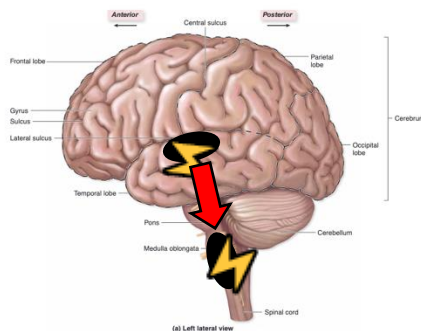
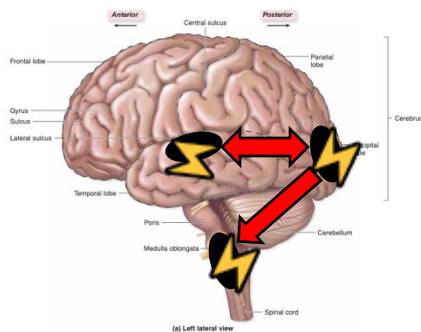
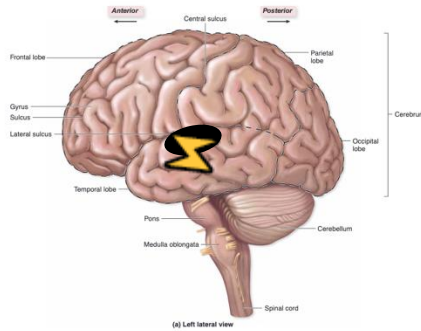
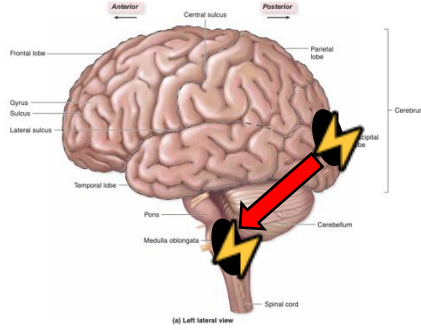
- Research suggests this is NOT the case
- R to CS does not *always* match R to US

In mice:

shock → jump

tone : shock

tone → freeze

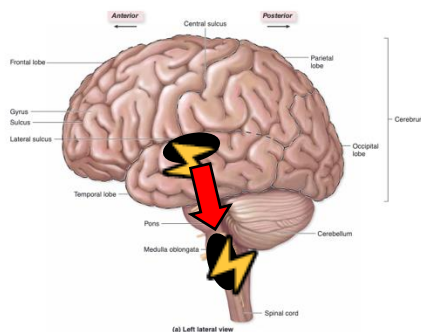
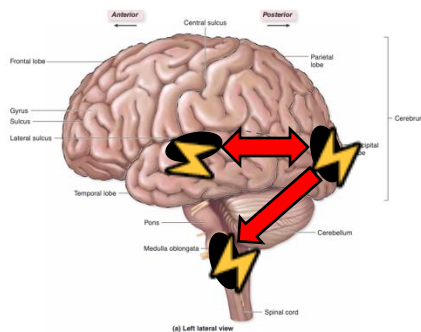
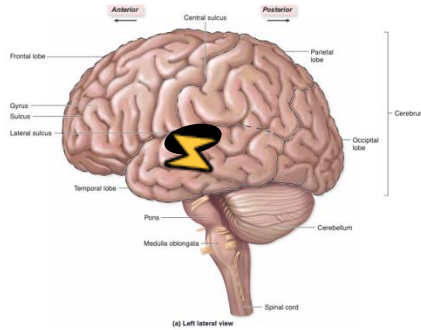
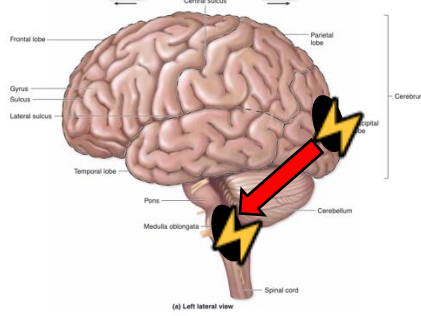


Theoretical Explanations of Classical Conditioning

Stimulus-Substitution Theory – CS acts as substitute for US in triggering neural activity

- Research suggests this is NOT the case
- CR does not *always* match US typical reflex

Maybe the CR is the body/brain preparing for the US (rather than *responding* to the CS)? (freeze = prepare for shock, not responding to the tone ?)



Theoretical Explanations of Classical Conditioning

Preparatory-Response Theory – R to CS
prepares organism for presentation of US

Theoretical Explanations of Classical Conditioning

Preparatory-Response Theory – R to CS
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Salivation $\neq$ CS replacing US
Salivation = preparing for US

Theoretical Explanations of Classical Conditioning

Preparatory-Response Theory – R to CS
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Shock → Jump

Tone → No Jump

Tone : Shock → Jump

Tone → Freeze

Freeze (prepare for shock) then shock → jump

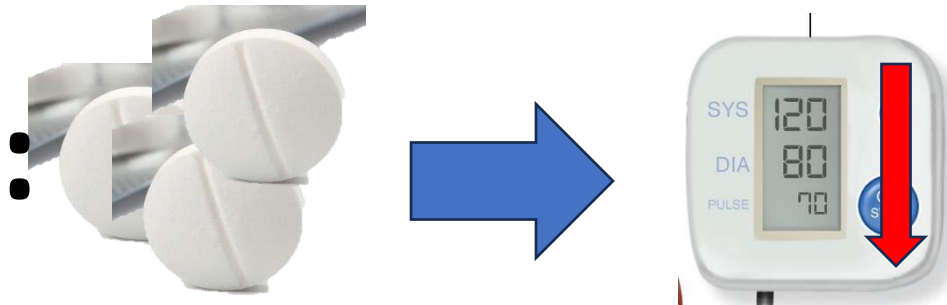
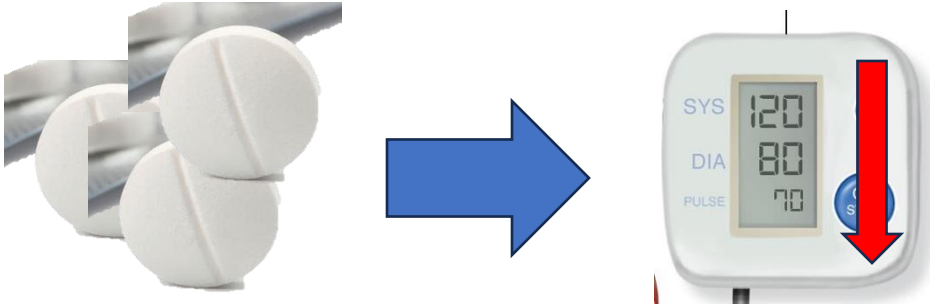
Theoretical Explanations of Classical Conditioning

Preparatory-Response Theory – R to CS
prepares organism for presentation of US

Salivation ≠ CS replacing US
Salivation = preparing for US

Becomes apparent in very *dangerous* scenarios...

Theoretical Explanations of Classical Conditioning



Preparatory-Response Theory – R to CS prepares organism for presentation of US

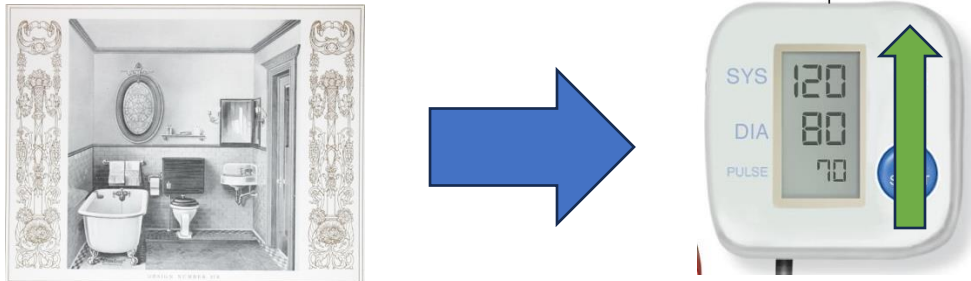
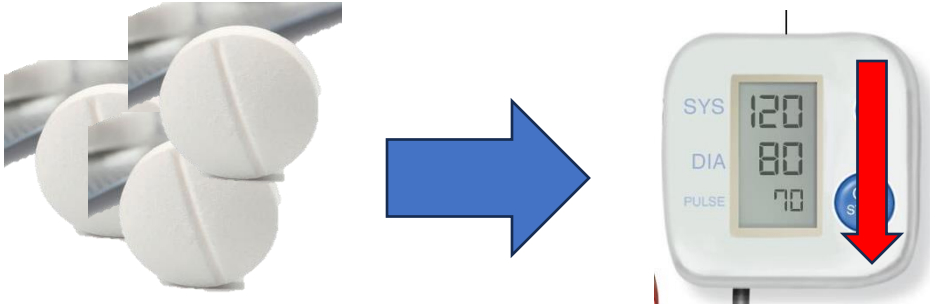
Analgesic → BP drop

Bathroom : Analgesic → BP drop

Bathroom → BP increase

Goal: **homeostasis**

Theoretical Explanations of Classical Conditioning



Preparatory-Response Theory – R to CS prepares organism for presentation of US

Analgesic → BP drop

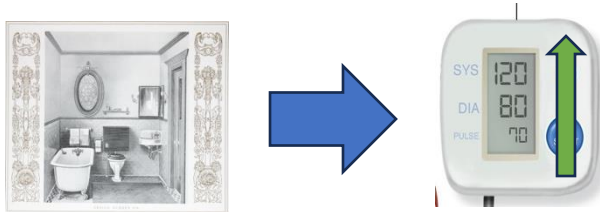
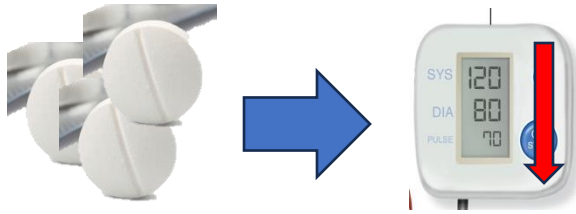
Bathroom : Analgesic → BP drop

Bathroom → BP increase

Goal: **homeostasis**

the regulation by an organism of all aspects of its internal environment (APA)

Theoretical Explanations of Classical Conditioning



Opponent-Process Theory – technically about emotions

Opponent-Process Theory is not *unique* to Preparatory-Response Theory, it just explains what we're seeing here extremely well

Preparatory-Response Theory – R to CS prepares organism for US presentation

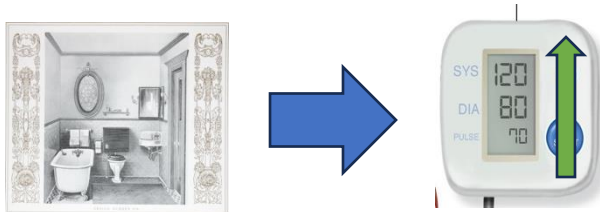
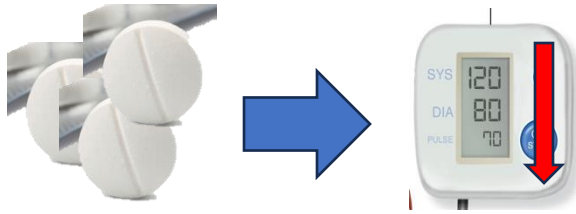
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Theoretical Explanations of Classical Conditioning



Opponent-Process Theory

initial reactions to emotional events or stimuli are followed by the opposite reaction over time or when that stimuli is removed (APA)

Preparatory-Response Theory – R to CS prepares organism for US presentation

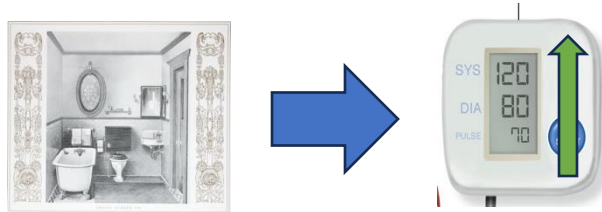
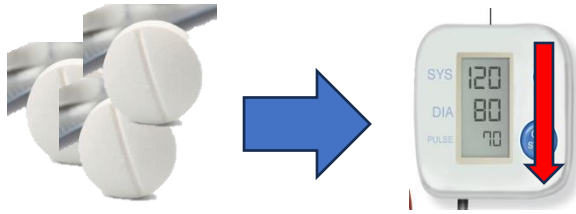
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Theoretical Explanations of Classical Conditioning



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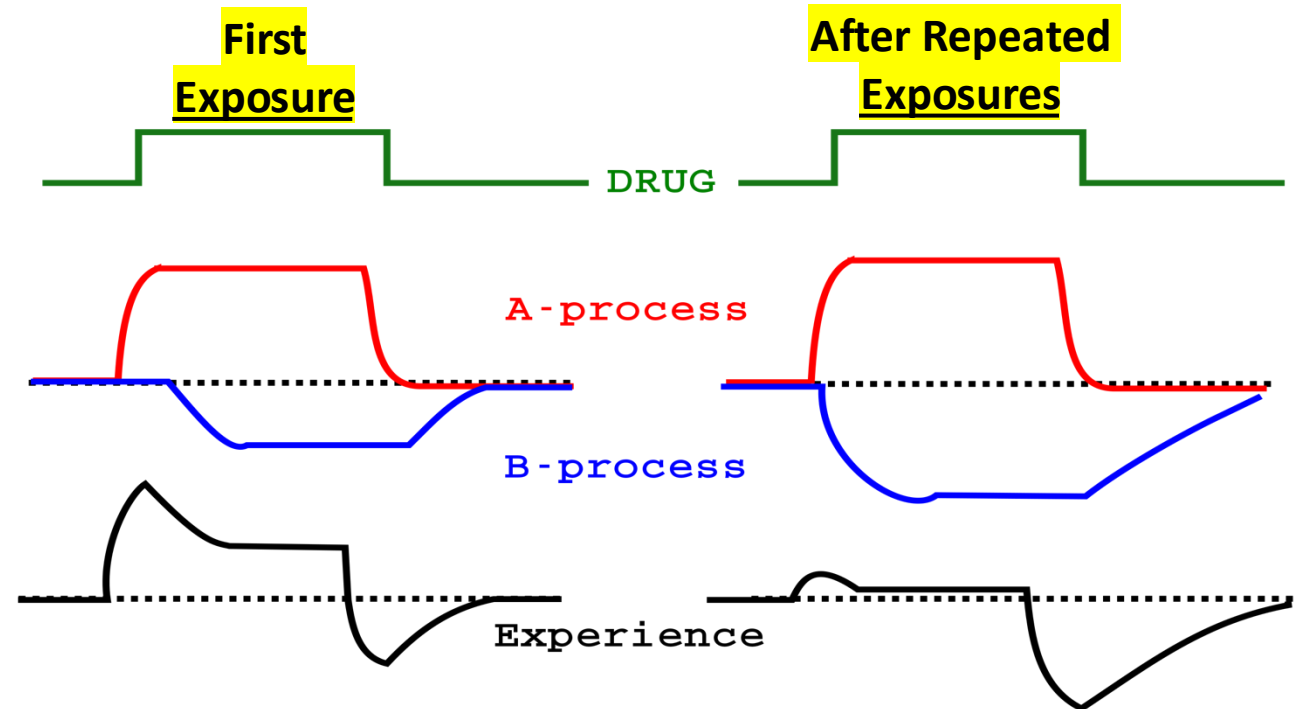
Bathroom → BP increase

Goal: **homeostasis**

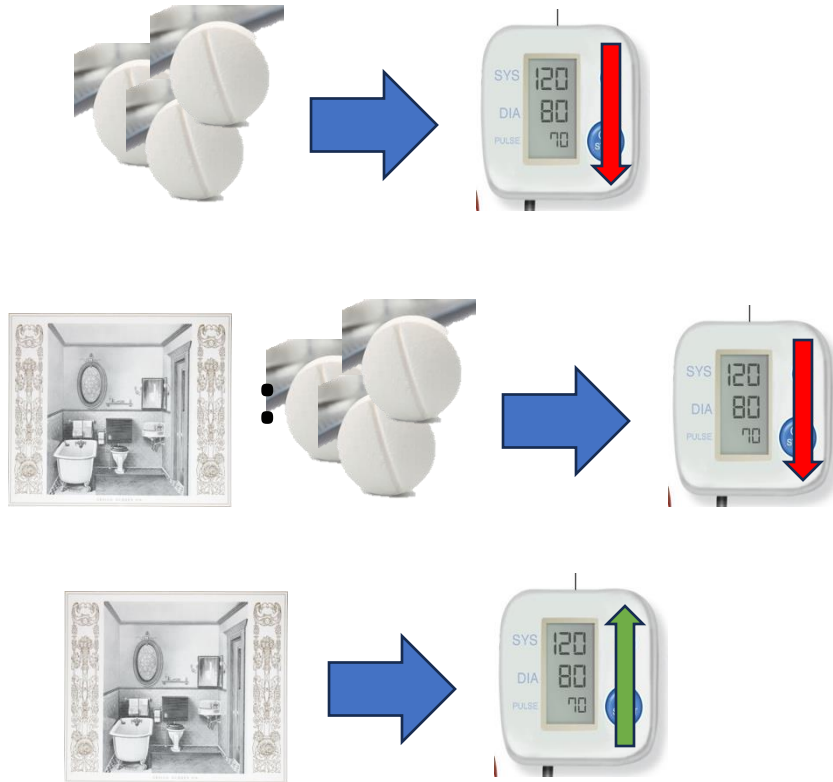
Opponent-Process Theory

First Exposure: Compensatory Process (“B” process) has slower onset & offset than natural response

After Repeat Exposure: Compensatory Process (“B” process) will quicken & strengthen with repeated exposure to stimulus



Theoretical Explanations of Classical Conditioning



Preparatory-Response Theory – R to CS prepares organism for US presentation

Analgesic → BP drop

Bathroom : Analgesic → BP drop

Bathroom → BP increase

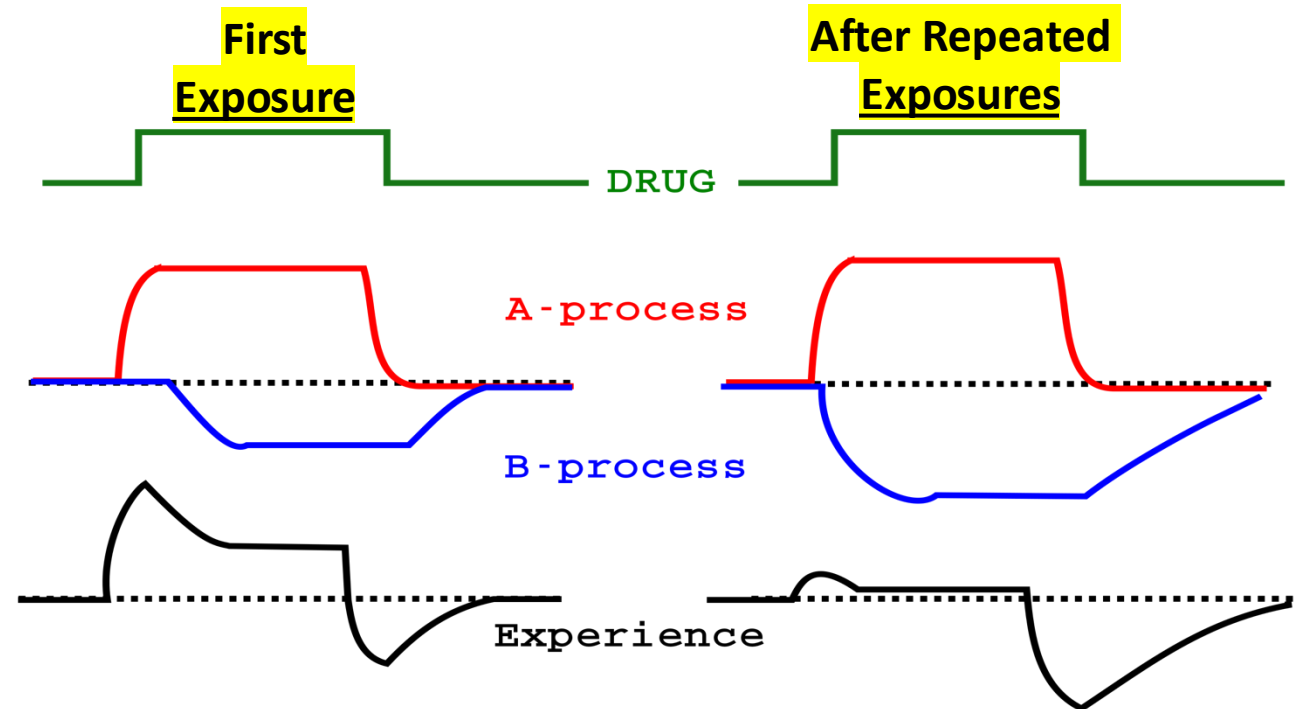
Goal: *homeostasis*

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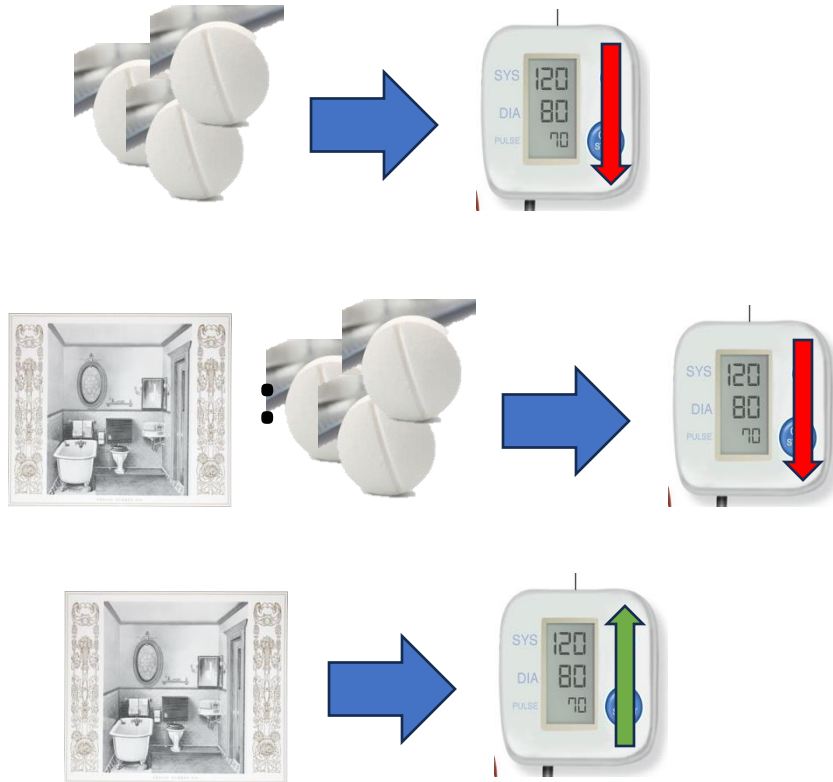
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Theoretical Explanations of Classical Conditioning



Preparatory-Response Theory – R to CS prepares organism for US presentation

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Goal: **homeostasis**

Opponent-Process Theory

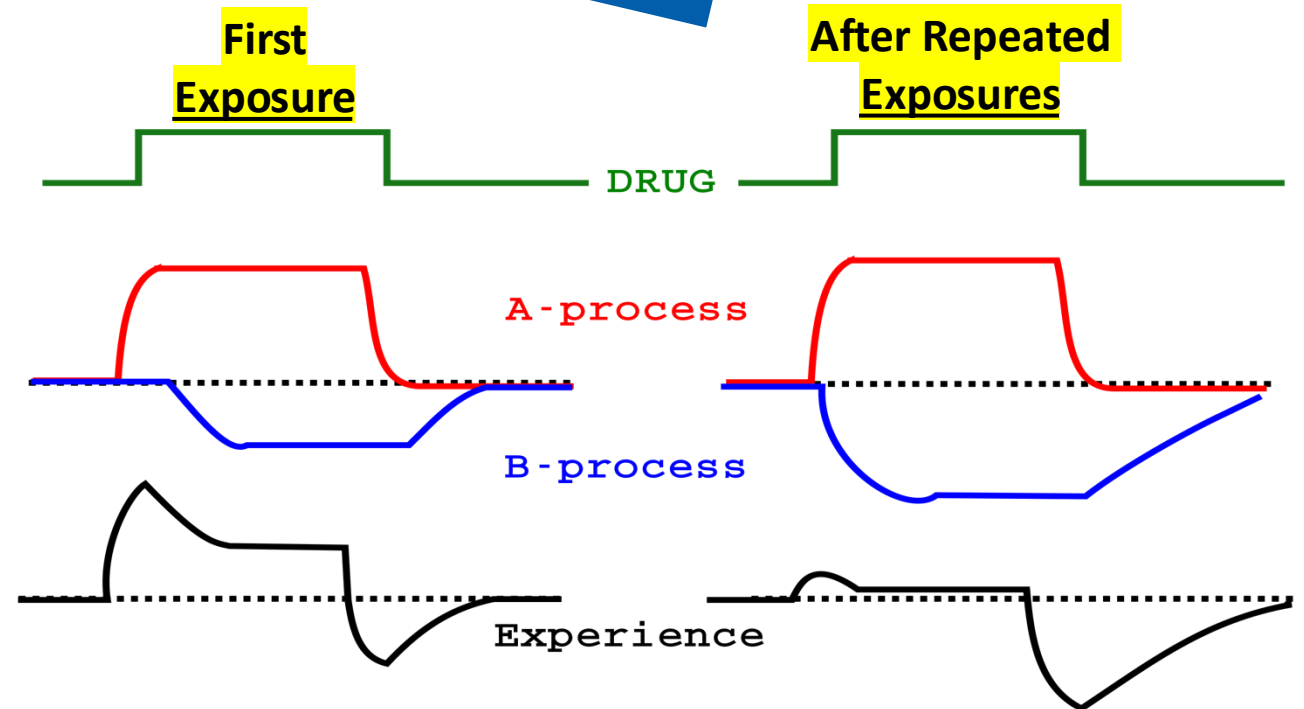
Which aspect of Classical Conditioning could be considered the “A” process and which the “B” process??

Sensory Process (“B” process) has an offset than natural response

Sensory Process (“B” process) with repeated exposure

us

homeostasis



Theoretical Explanations of Classical Conditioning

Opponent-Process Theory

First Exposure: Compensatory Process ("B" process) has slower onset & offset than natural response

After Repeat Exposure: Compensatory Process ("B" process) will quicken & strengthen with repeated exposure to stimulus

Goal: *homeostasis*

"A" Process



"B" Process



First Exposure

After Repeated Exposures

DRUG

A-process

B-process

Experience

Preparatory-Response Theory – R to CS prepares organism for US presentation

Analgesic → BP drop

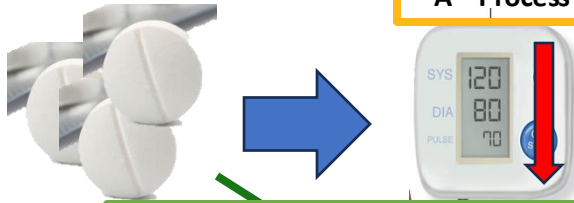
Bathroom : Analgesic → BP drop

Bathroom → BP increase

Goal: *homeostasis*

Theoretical Explanations of Classical Conditioning

"A" Process



Using Opponent Process to help explain Preparatory-Response:

"B" Process (response to CS)
prepares organism for expected **"A" Process (response to US)**



Preparatory-Response Theory – R to CS prepares organism for US presentation

Analgesic → BP drop

Bathroom : Analgesic → BP drop

Bathroom → BP increase

Goal: *homeostasis*

Opponent-Process Theory

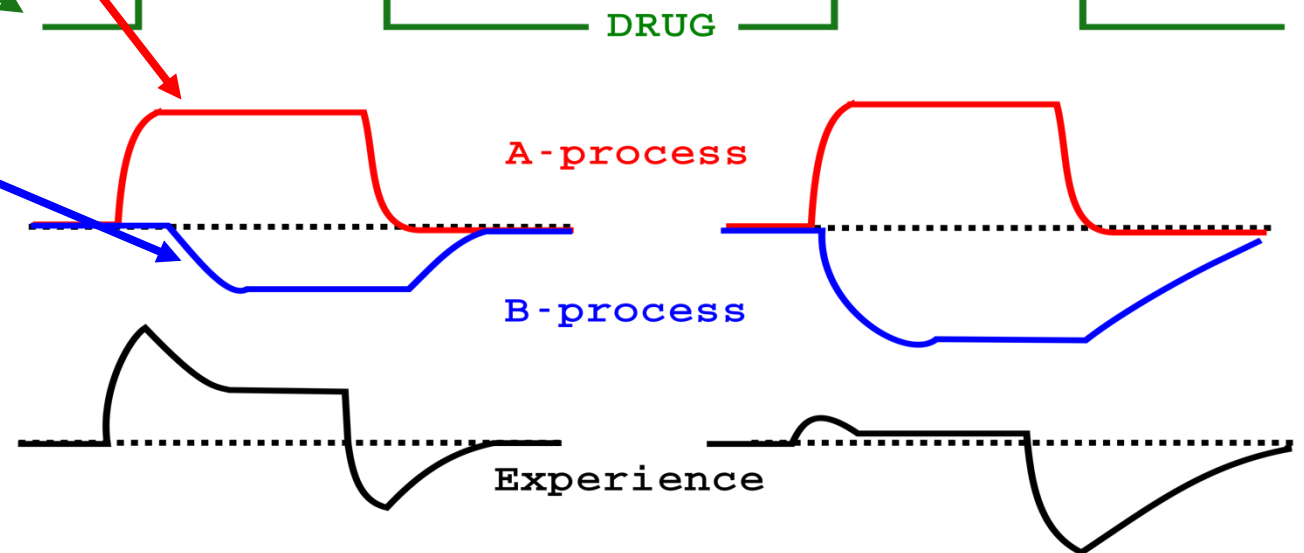
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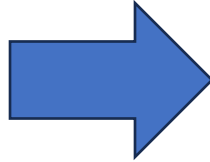
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First Exposure

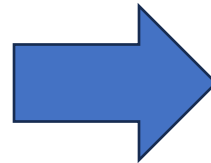
After Repeated Exposures



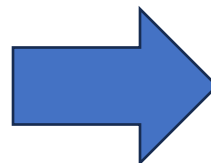
Theoretical Explanations of Classical Conditioning



"A" Process



"B" Process



What are some other examples of scenarios where we might see P-R Model in action?

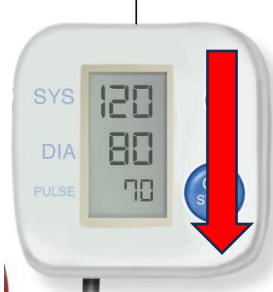
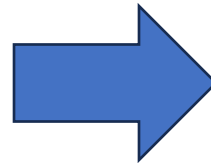
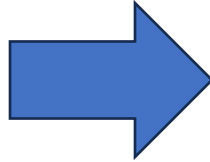
Preparatory-Response Theory – R to CS prepares organism for presentation of US

Some scenarios where this pattern is obvious:

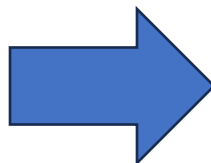
Theoretical Explanations of Classical Condition



"A" Process



"B" Process



How might P-R Theory show up in terms of cravings or tolerances to a drug/alcohol?

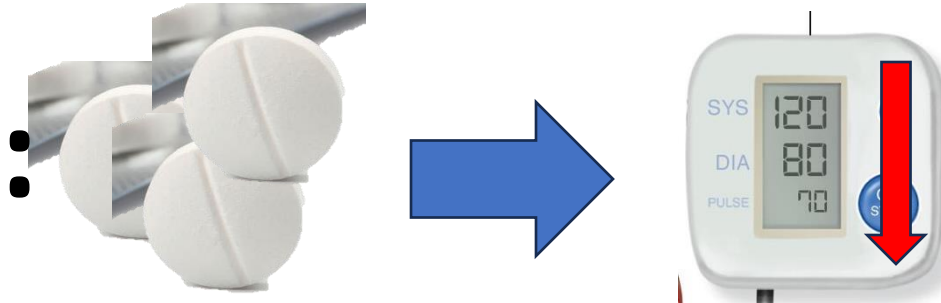
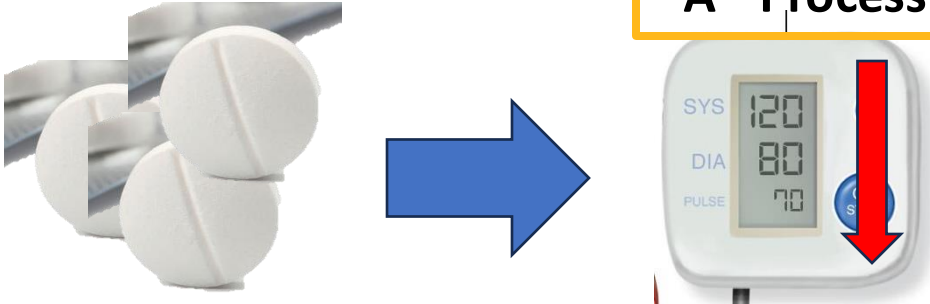
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Theoretical Explanations of Classical Conditioning

"A" Process



"B" Process



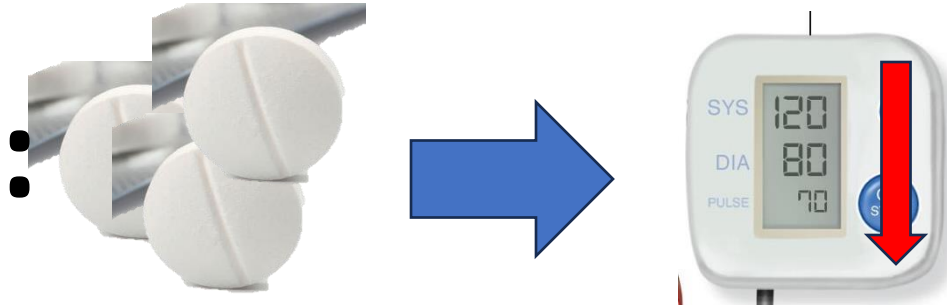
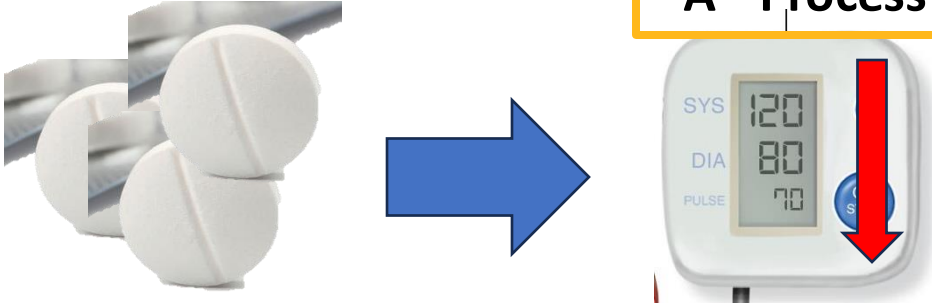
Preparatory-Response Theory – R to CS prepares organism for presentation of US

Some scenarios where this pattern is obvious:

- Cravings
 - Alcoholic
 - Always drinks at Joe's house
 - Goes to rehab
 - No more cravings when around alcohol
 - Goes to Joe's house
 - "B" process occurs again
 - Interprets this as a craving
- Tolerances

Theoretical Explanations of Classical Conditioning

"A" Process



"B" Process



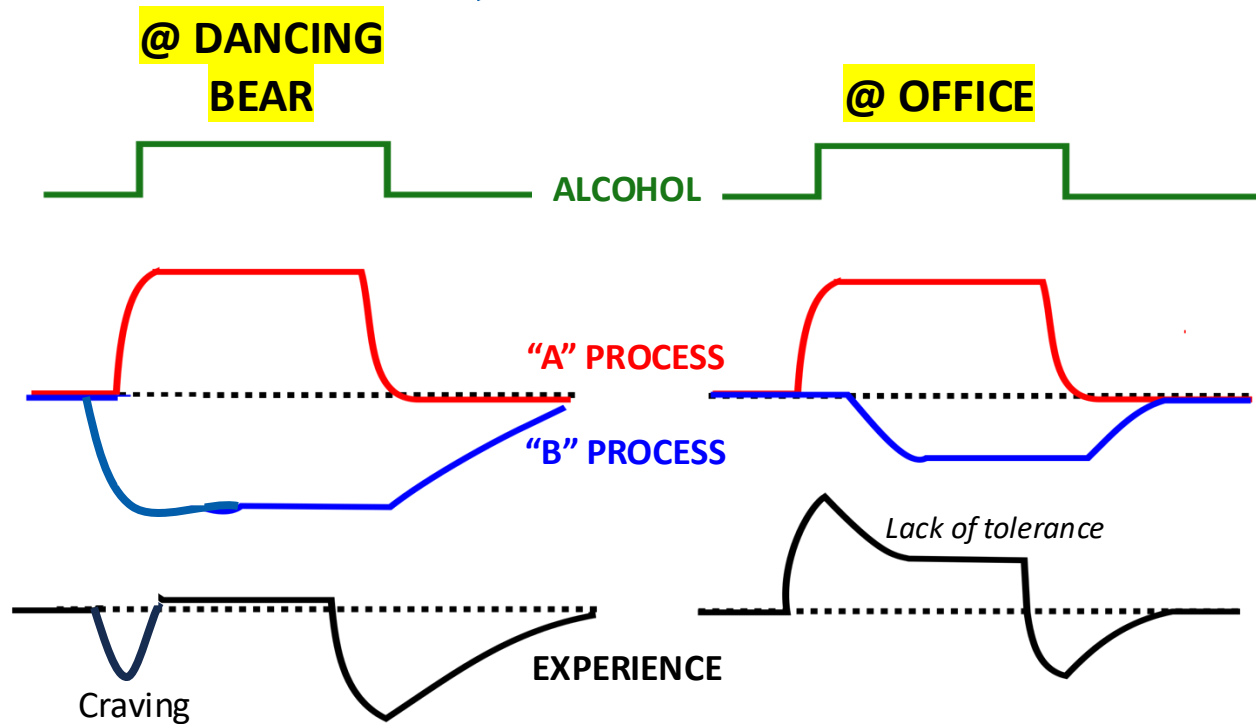
Preparatory-Response Theory – R to CS prepares organism for presentation of US

Some scenarios where this pattern is obvious:

- Cravings
- Tolerances
 - Casual drinker, ~3 drinks on a given night
 - Always drinks at Dancing Bear
 - Goes to party at office
 - 3 drinks, like normal
 - Much more intoxicated than normal

What's this theory/model called again?

Explanations of Classical Conditioning

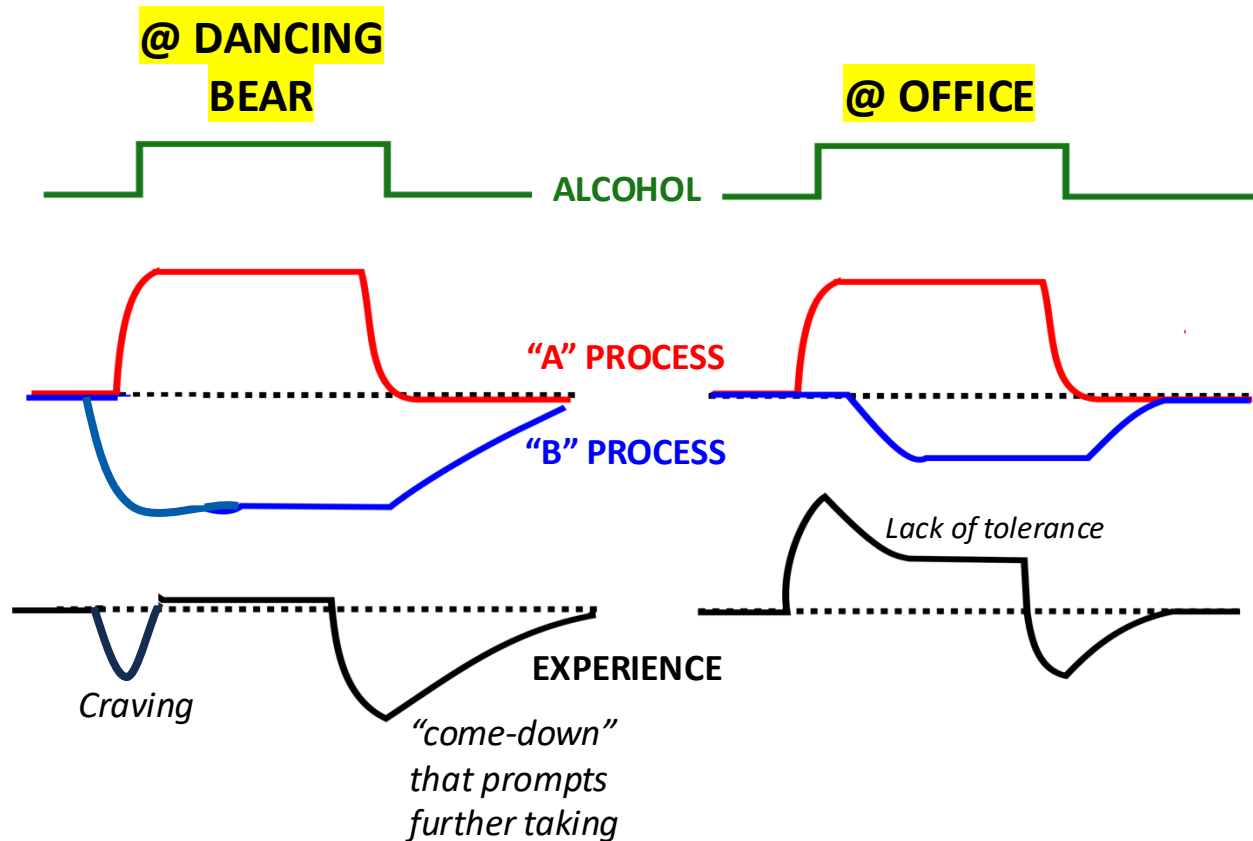


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Theoretical Explanations of Classical Conditioning

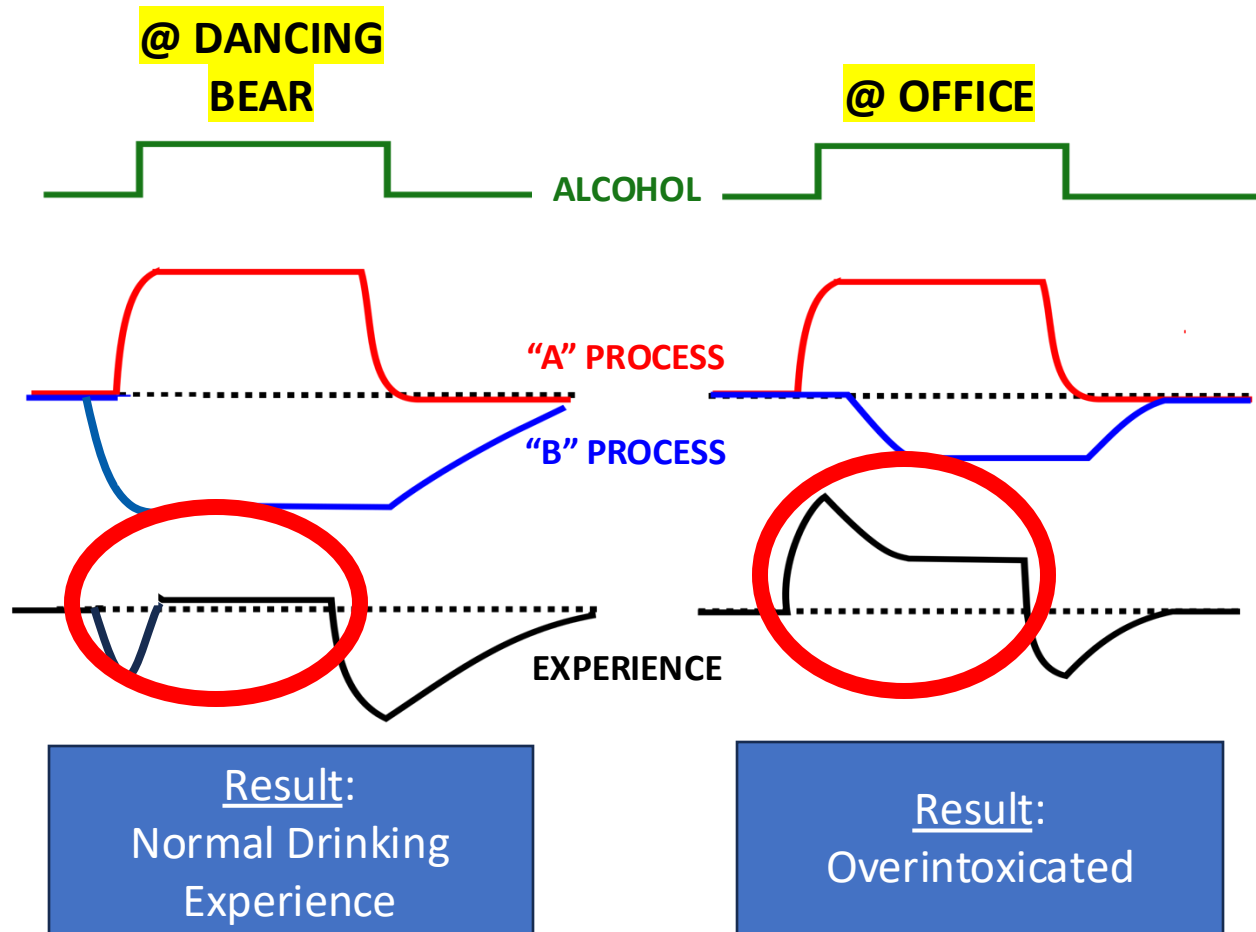


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Theoretical Explanations of Classical Conditioning



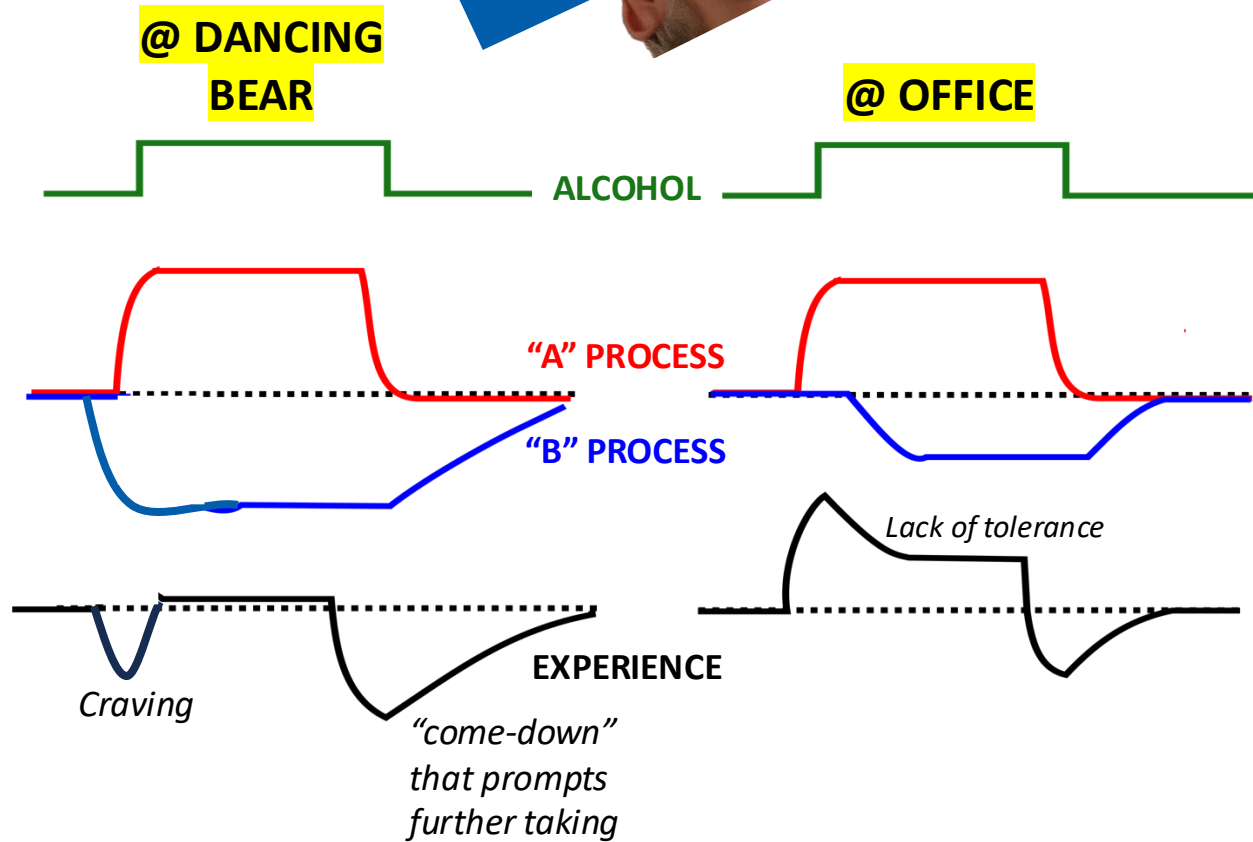
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Theory: Explanations of Classical Conditioning

What term might we call Dancing Bear in a classical conditioning paradigm?



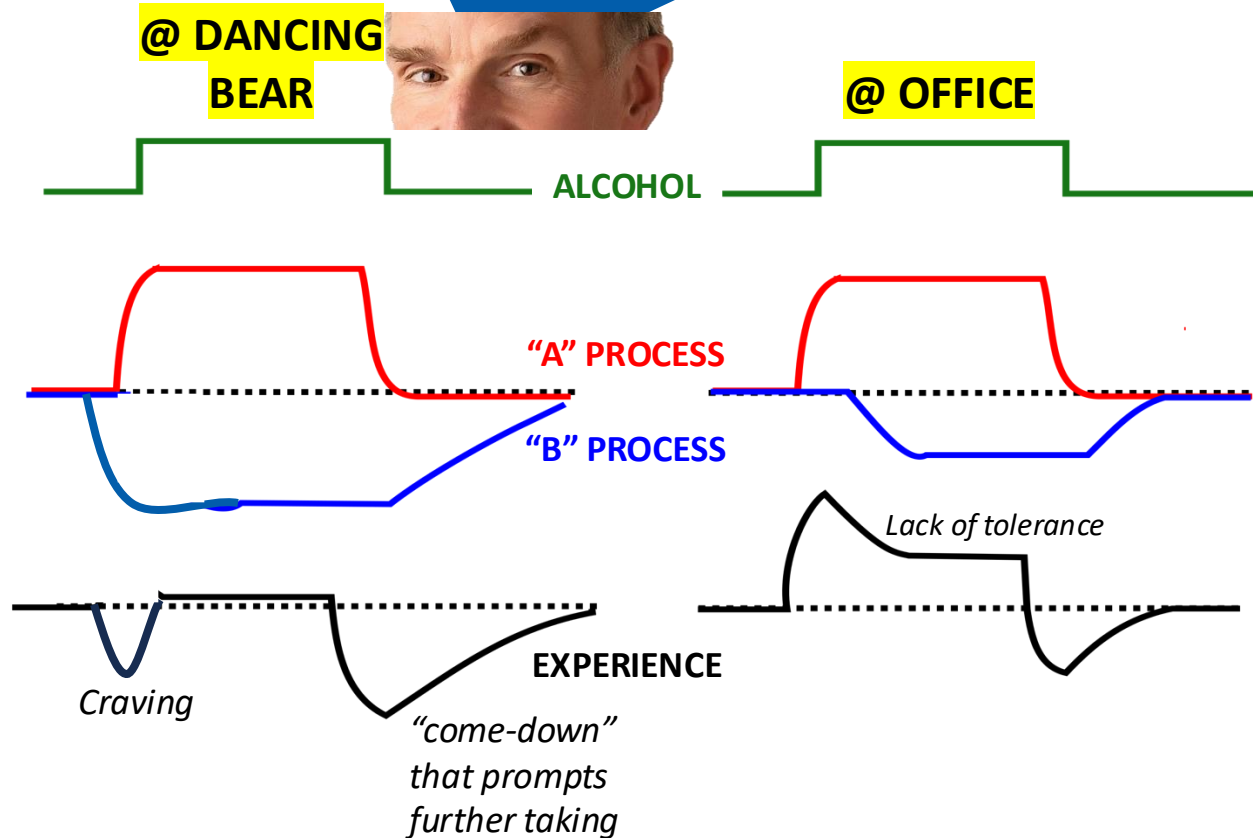
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Theoretical Foundations of Classical Conditioning

How would we write out that paradigm in shorthand?



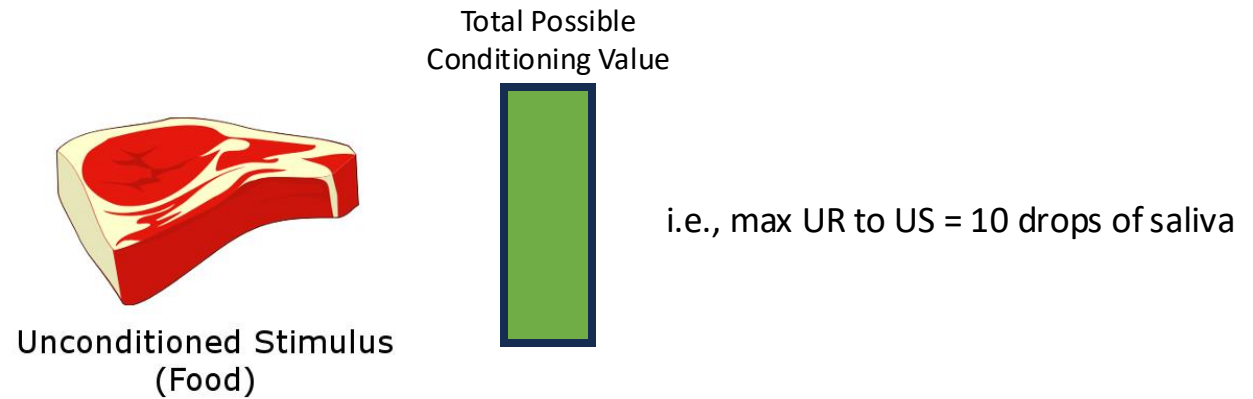
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Theoretical Explanations of Classical Conditioning

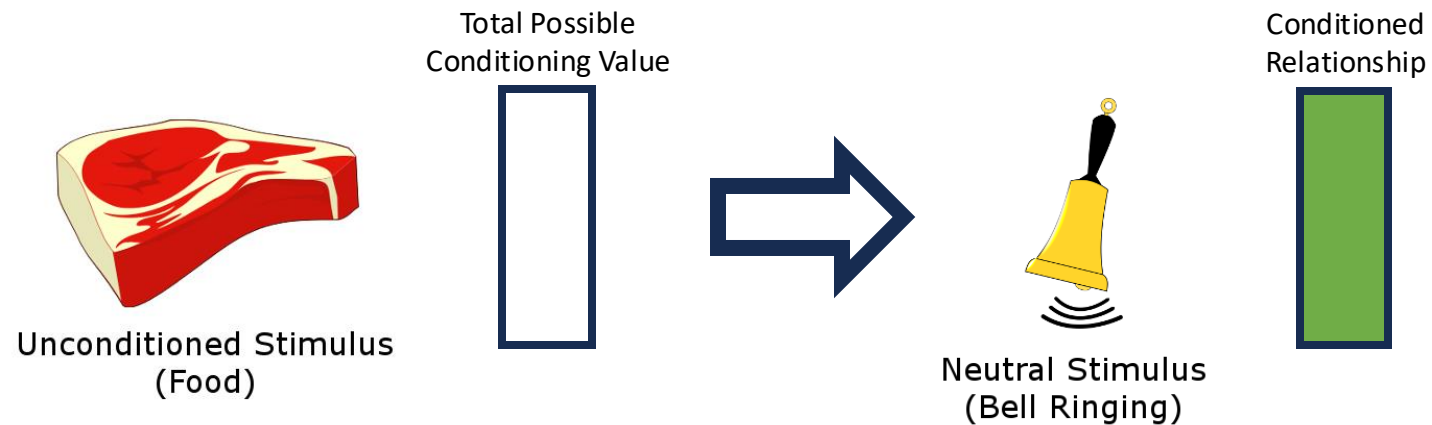
Rescorla-Wagner Theory – A given US can support only so much conditioning, which must be distributed across various present CS's



Theoretical Explanations of Classical Conditioning

Rescorla-Wagner Theory – A given US can support only so much conditioning, which must be distributed across various present CS's

Bell : Steak → Salivation



*Maximum UR to US was 10 drops of saliva
So maximum CR to CS = 10 drops of saliva*

Theoretical Explanations of Classical Conditioning

Rescorla-Wagner Theory – A given US can support only so much conditioning, which must be distributed across various present CS's

Value of US can differ, which affects CS-CR relationship

Less amount



Unconditioned Stimulus
(Food)

Total Possible
Conditioning Value

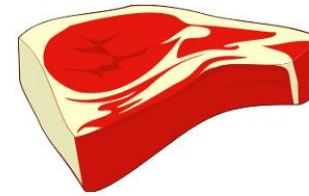


Less preferred



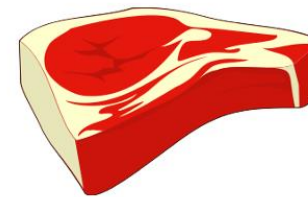
Unconditioned Stimulus
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Total Possible
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Unconditioned Stimulus
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Total Possible
Conditioning Value

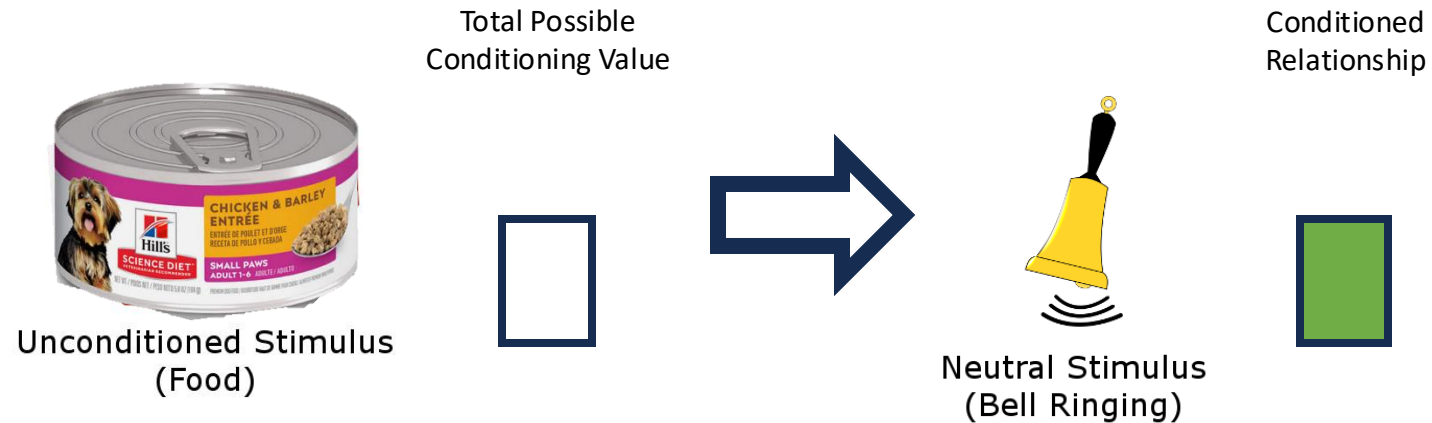


Theoretical Explanations of Classical Conditioning

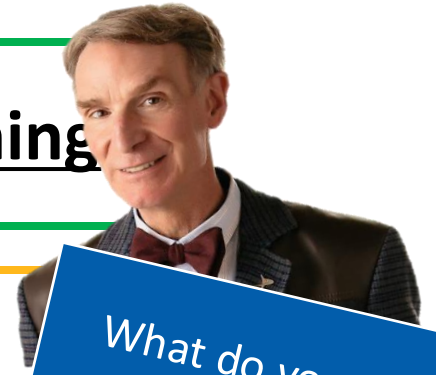
Rescorla-Wagner Theory – A given US can support only so much conditioning, which must be distributed across various present CS's

Value of US can differ, which affects CS-CR relationship

Bell : Canned Food → Salivation



Theoretical Explanations of Classical Conditioning

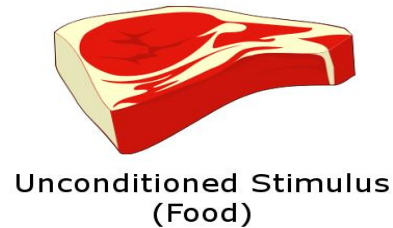


Rescorla-Wagner Theory – A given US can support only so much conditioning, which must be distributed across various present CS's

Value of US can differ, which affects CS-CR relationship

What do you think will happen here?

Bell : Steak → Salivation



Total Possible
Conditioning Value



Neutral Stimulus
(Bell Ringing)

Conditioned
Relationship



Later, present less of US



Unconditioned Stimulus
(Food)

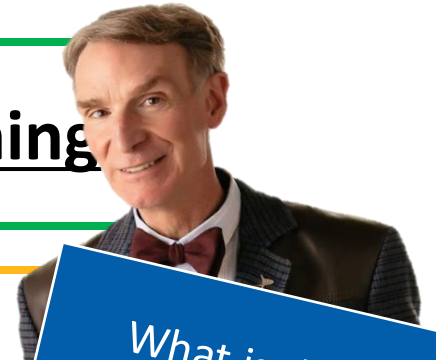


Then, re-present CS



Neutral Stimulus
(Bell Ringing)

Theoretical Explanations of Classical Conditioning

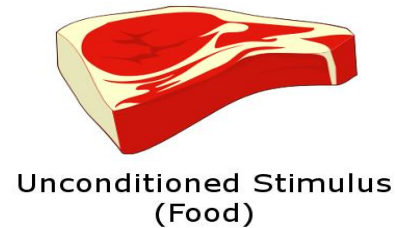


Rescorla-Wagner Theory – A given US can support only so much conditioning, which must be distributed across various present CS's

Value of US can differ, which affects CS-CR relationship

What is this process called?

Bell : Steak → Salivation



Unconditioned Stimulus
(Food)

Total Possible
Conditioning Value



Neutral Stimulus
(Bell Ringing)

Conditioned
Relationship



Later, present less of US



Unconditioned Stimulus
(Food)



Then, re-present CS



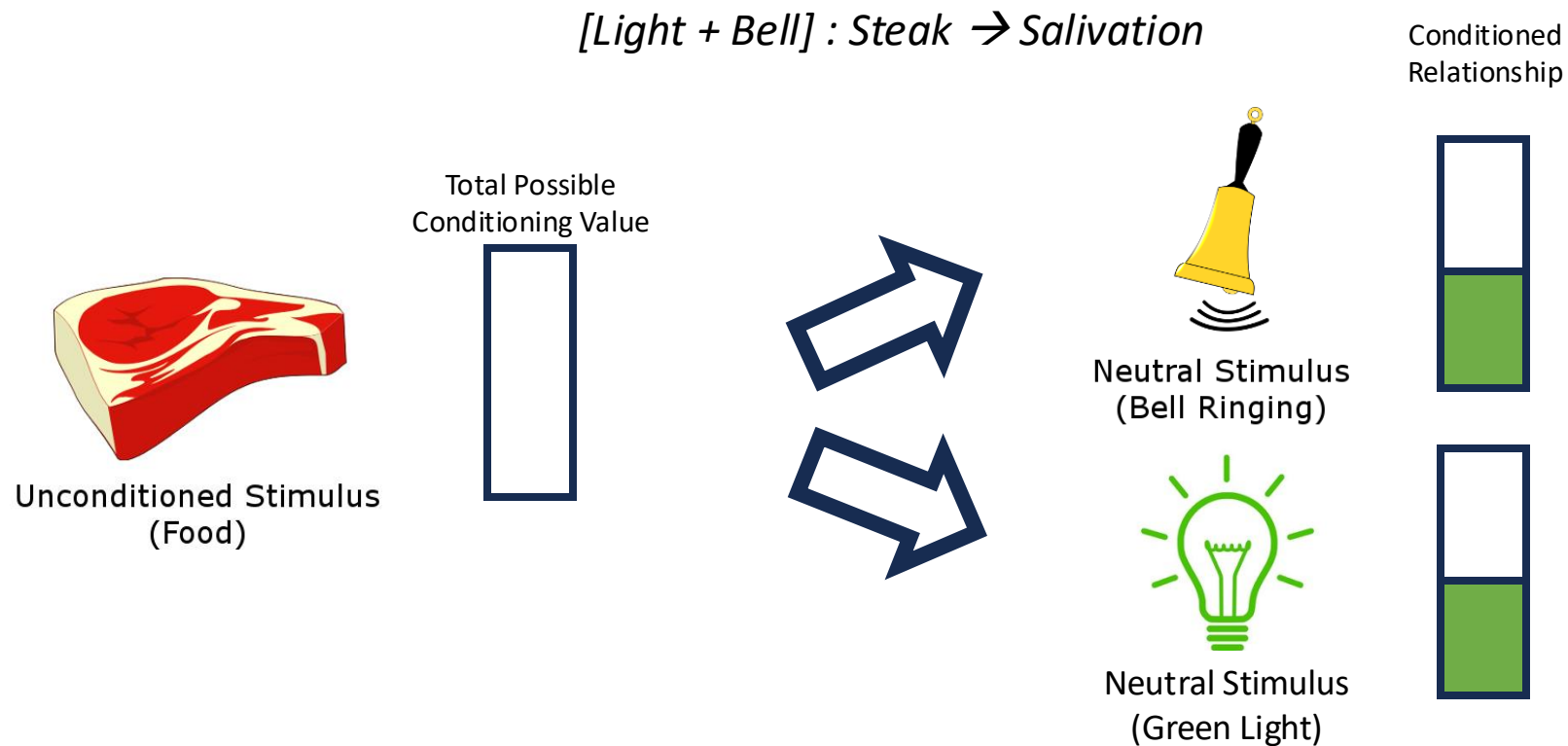
Neutral Stimulus
(Bell Ringing)



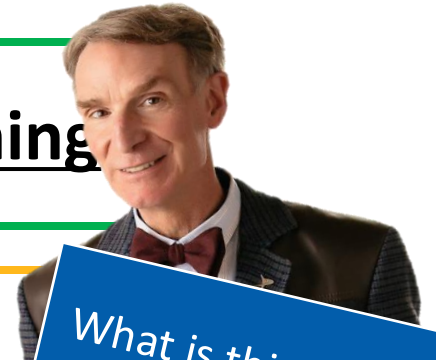
Theoretical Explanations of Classical Conditioning

Rescorla-Wagner Theory – A given US can support only so much conditioning, which must be distributed across various present CS's

If two CS are involved, they must compete for the conditioning



Theoretical Explanations of Classical Conditioning

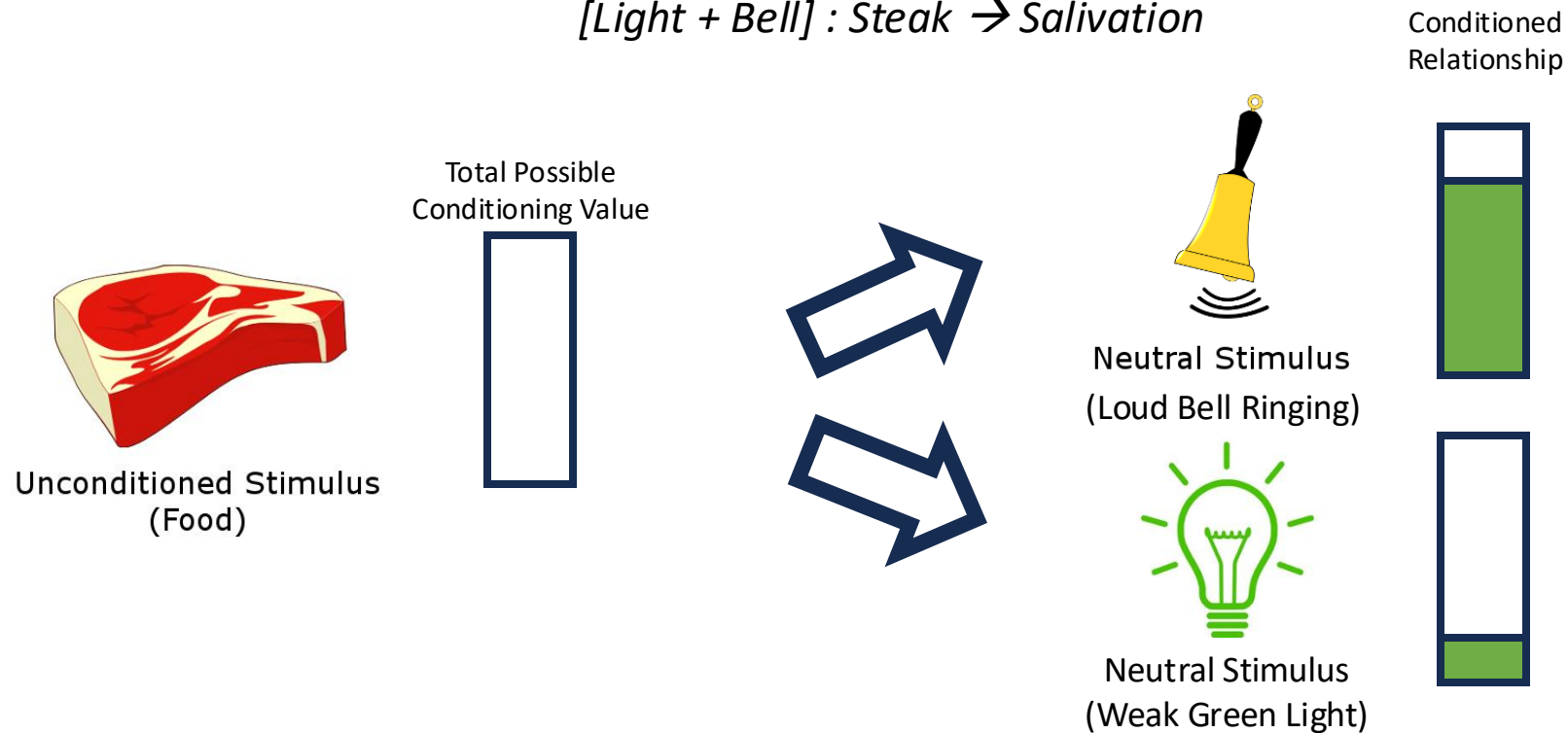


Rescorla-Wagner Theory – A given US can support only so much conditioning, which must be distributed across various present CS's

If one of CS are stronger, ... may occur in this competition

What is this phenomenon called...?

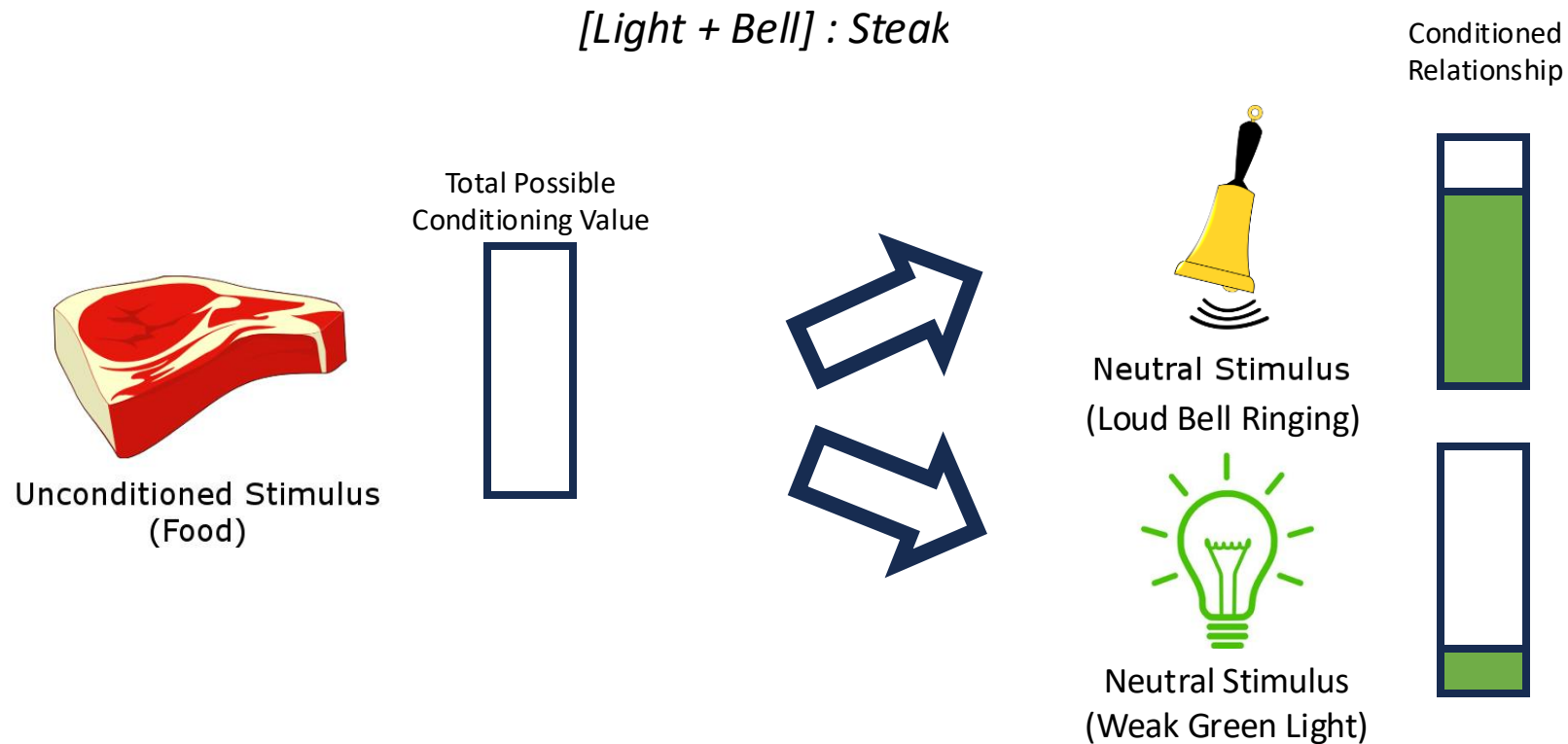
[Light + Bell] : Steak → Salivation



Theoretical Explanations of Classical Conditioning

Rescorla-Wagner Theory – A given US can support only so much conditioning, which must be distributed across various present CS's

If one of CS are stronger, **overshadowing** may occur in this competition



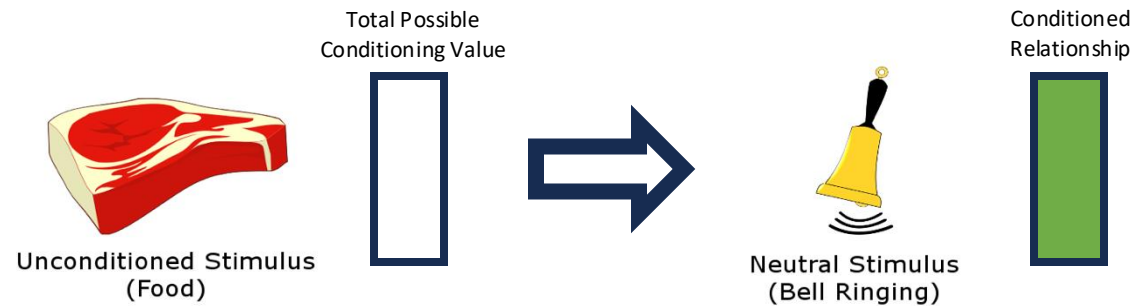
Theoretical Explanations of Classical Conditioning

Rescorla-Wagner Theory – A given US can support only so much conditioning, which must be distributed across various present CS's

If CS already has full US conditioning value, new compounded CS have nothing to compete for (...)

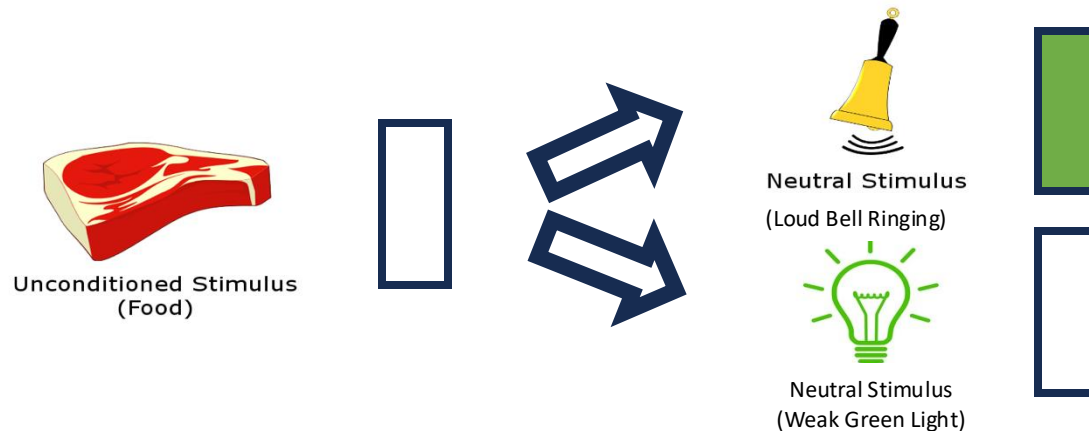
What is this phenomenon called...?

Bell : Steak → Salivation



Later on...

[Light + Bell] : Steak → Salivation

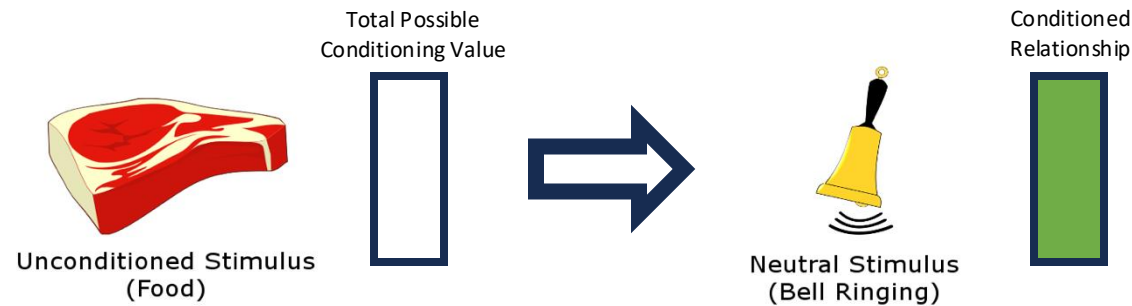


Theoretical Explanations of Classical Conditioning

Rescorla-Wagner Theory – A given US can support only so much conditioning, which must be distributed across various present CS's

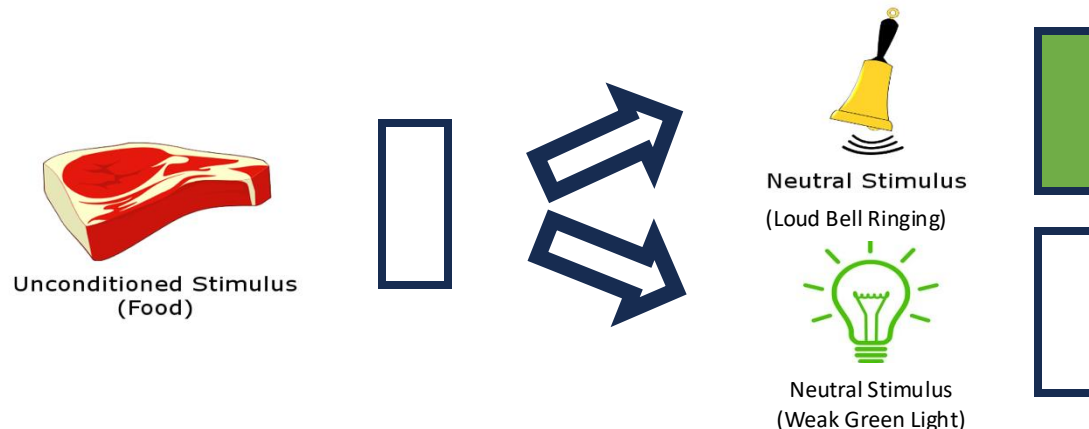
If CS already has full US conditioning value, new compounded CS have nothing to compete for (**blocking**)

Bell : Steak → Salivation



Later on...

[Light + Bell] : Steak → Salivation



Theoretical Explanations of Classical Conditioning

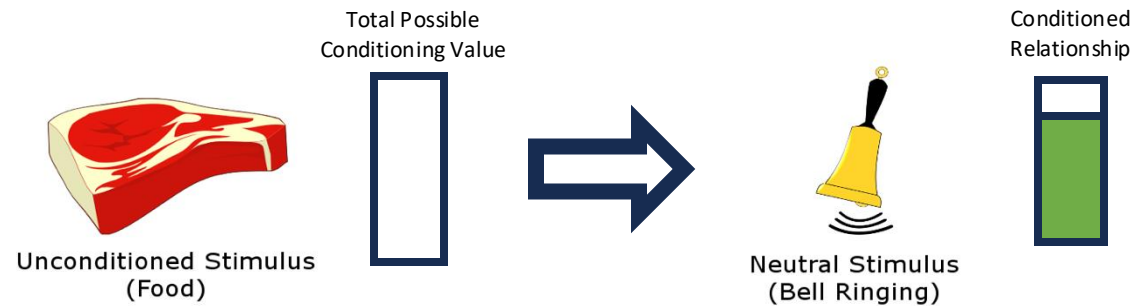


What's happening here?

Rescorla-Wagner Theory – A given US can support only so much conditioning, which must be distributed across various present CS's

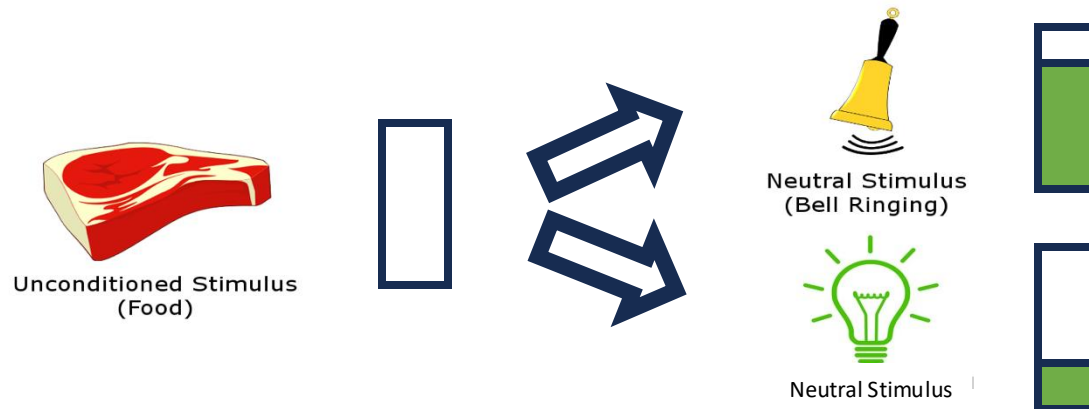
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Later on...

[Light + Bell] : Steak → Salivation



Theoretical Explanations of Classical Conditioning

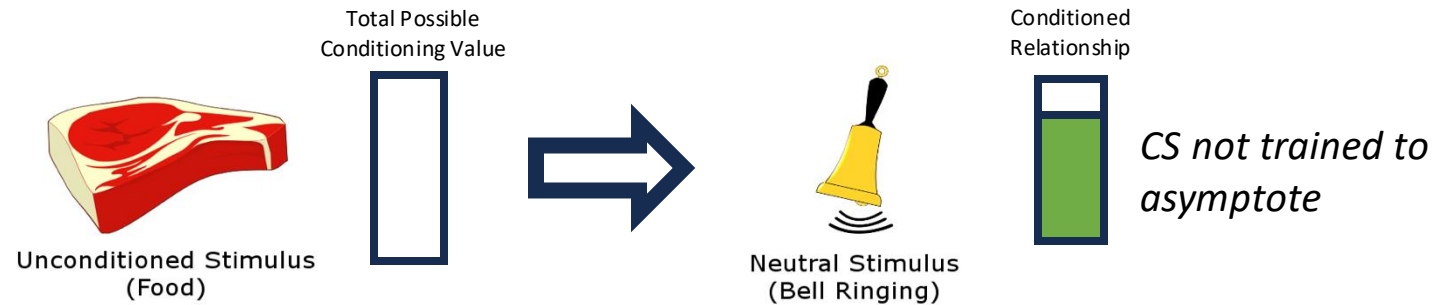


What's happening here?

Rescorla-Wagner Theory – A given US can support only so much conditioning, which must be distributed across various present CS's

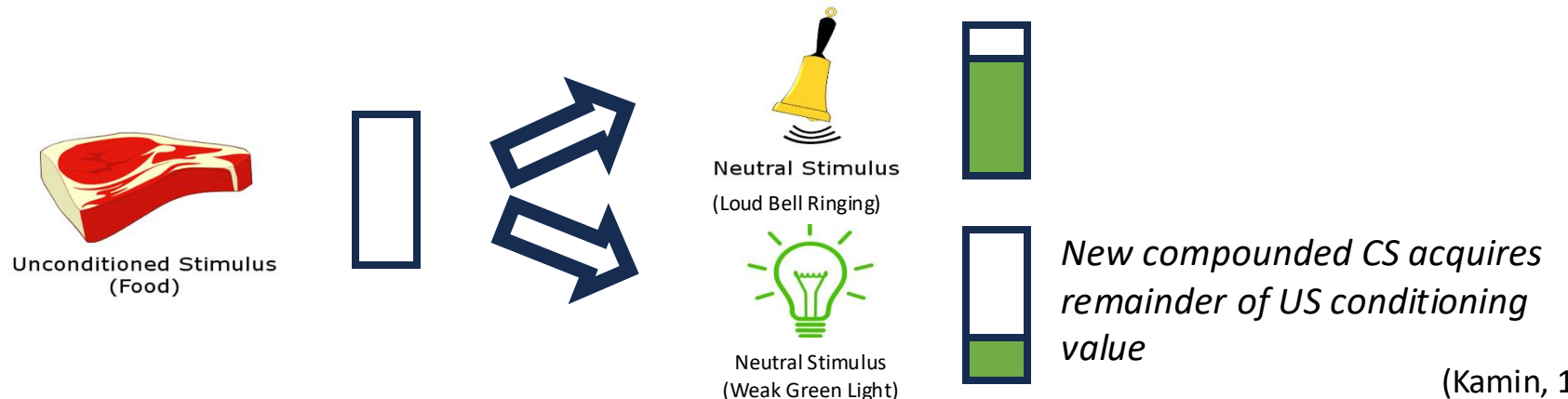
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Bell : Steak → Salivation



Later on...

[Light + Bell] : Steak → Salivation



(Kamin, 1968)

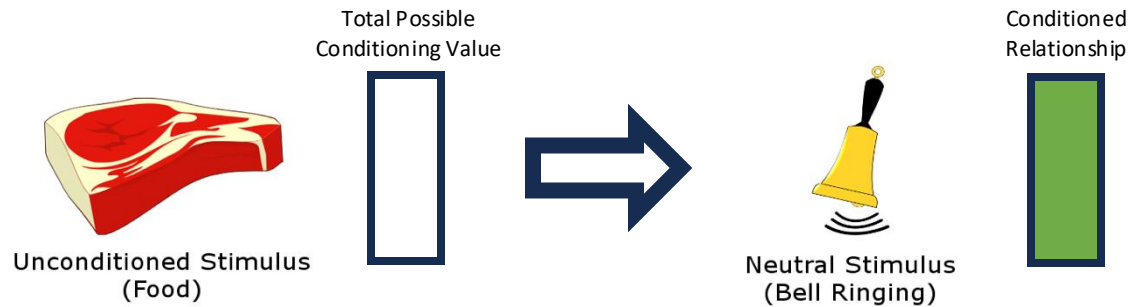
Theoretical Explanations of Classical Conditioning

Rescorla-Wagner Theory – A given US can support only so much conditioning, which must be distributed across various present CS's

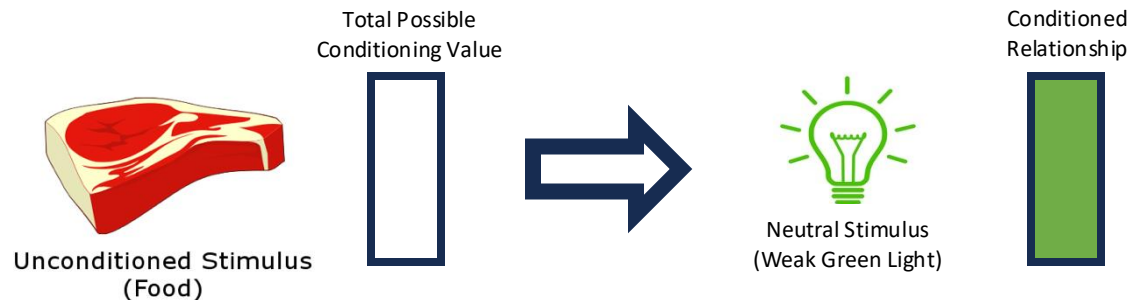
If we condition two stimuli independently, then compound them...

What do you think will happen here??

Bell : Steak → Salivation

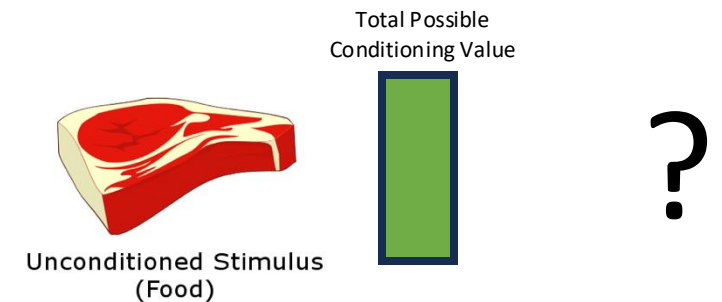


Light : Steak → Salivation



Later on...

[Light + Bell] : Steak → Salivation

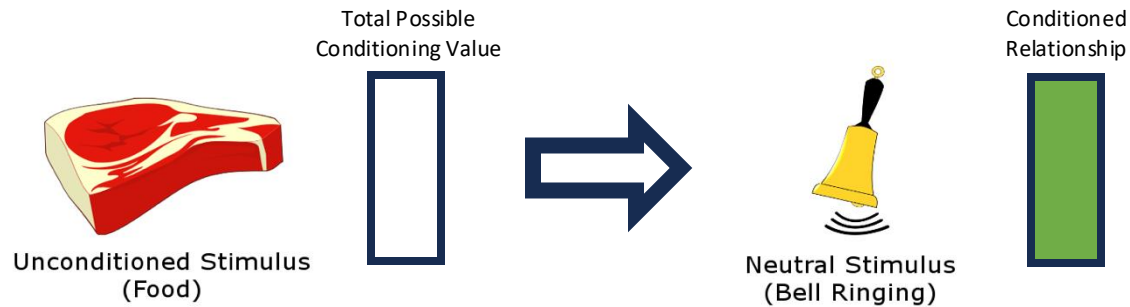


Theoretical Explanations of Classical Conditioning

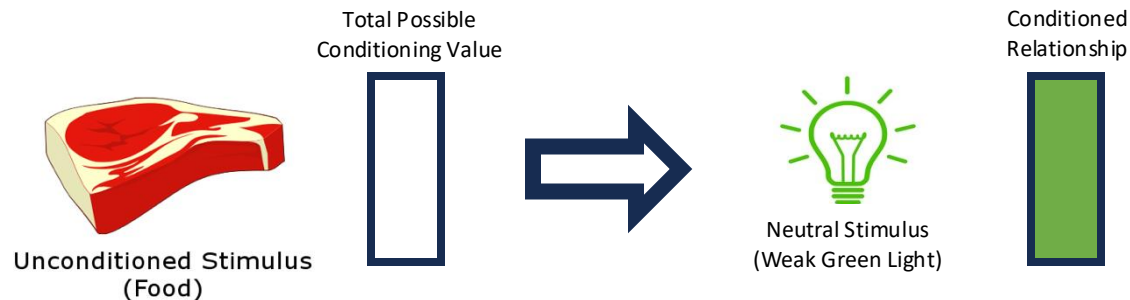
Rescorla-Wagner Theory – A given US can support only so much conditioning, which must be distributed across various present CS's

Overexpectation Effect – both R decrease to meet max value R of US
Still must compete for US' conditioning value

Bell : Steak → Salivation

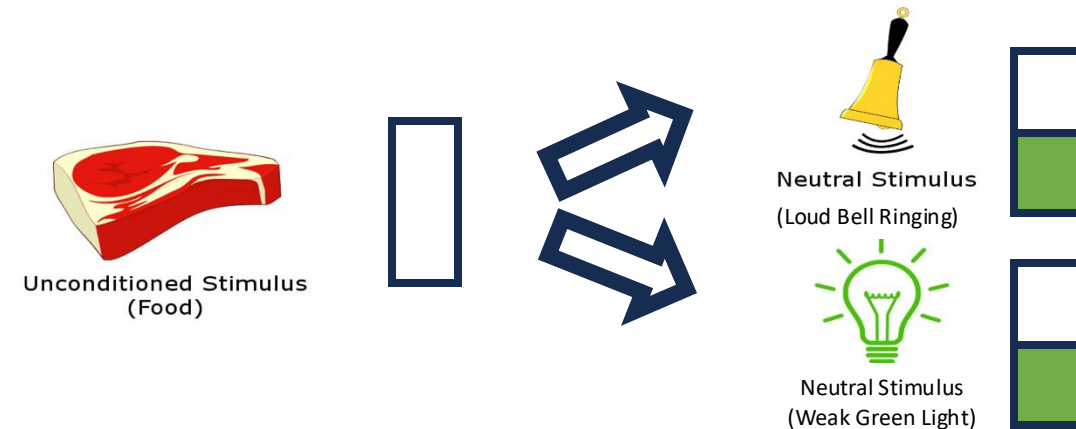


Light : Steak → Salivation



Later on...

[Light + Bell] : Steak → Salivation

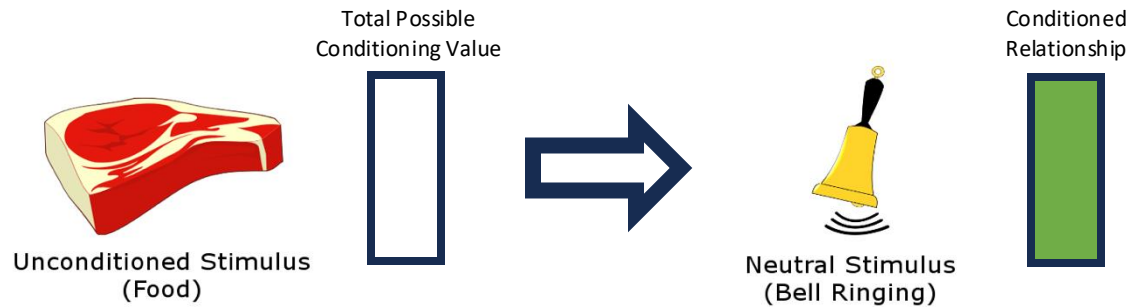


Theoretical Explanations of Classical Conditioning

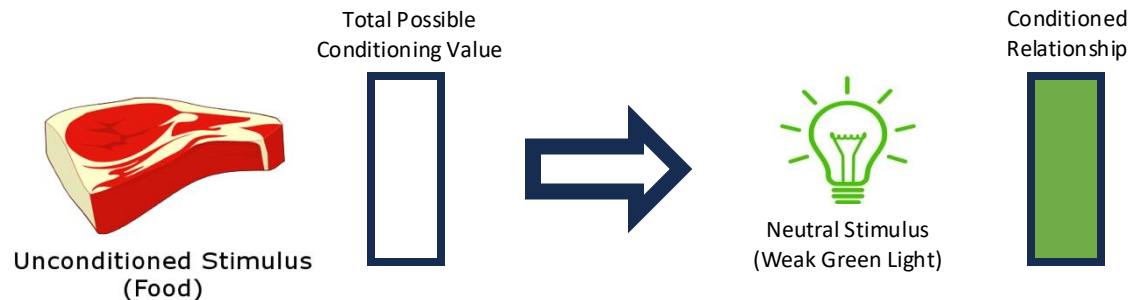
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Bell : Steak → Salivation



Light : Steak → Salivation



Later on...

[Light + Bell] : Steak → Salivation



Each CS will decrease its response strength after compound conditioning

Questions?